4.0. Lists 101

A list is a way to store data. It's similar to an array, but an array from a language such as Java is considerably more limited. Some key features of a list:

- It can store data
- The data doesn't have to be of the same type
- A list doesn't have a fixed size (it can be expanded or compressed)
- A list can be declared with 0, or some number of elements in it.

A list is extremely flexible as a result of these points. Declaring a list is simple.

```
#An empty list

alist = []

#A list with stuff in it

another_list = [1, 4, 7, 4.0, 'hello world']
```

To access an element of a list, use the following syntax:

```
nums = [4, 5, 13, 55]

#print out the variable at index 1 (the second item)

print (nums[1])
```

```
#output

5
```

4.1. List Functions

Lists are objects in Python. This means that there are some functions available to be called on lists. For the purposes of this tutorial we will cover a few of the more basic ones (for a full list of all the available functions see http://docs.python.org/2/tutorial/datastructures.html).

1. Adding an item to a list.
   list.append(x) adds an element x to the end of a list. Again, type doesn't actually matter (it can be a string, int… etc). Note: lists can be printed, as is done here.

```
nums = [3, 5, 6]

#Add a number to the end

nums.append(9)

print(nums)
```

```
#Lists print in the following format

[3, 5, 6, 9]
```

2. Inserting an item in a list
   list.insert() inserts an element x at index i. index i must be an integer. Remember that the computer starts counting at 0. For example:

```
nums = [3, 5, 6]
#Insert a number at index 1
nums.insert(1, 66)
print(nums)
```

```
[3, 66, 5, 6]
```

3. Removing an item from a list
   list.pop([i]) removes the item at the given index *i*, and returns it. Why are there [] around the parameter *i*? Because it's optional! If the parameter *i* is omitted then the last item from the list is removed and returned.

```
nums = [33, 567, 123, 35435, 4]
#get and remove the value at index 3
j = nums.pop(3)
print(j)
print(nums)
```

```
35435

[33, 567, 123, 4] #Note that 35435 is gone
```

4. Sorting a list
   list.sort() puts a list in order from smallest to largest. list.reverse() will put the list in reverse order. For example :

```
nums = [3, 5, 6, 3, 2, 1, 9, 11, 3, 4,3]
#sort the numbers
nums.sort()
print(nums)
# put the list in reverse order
nums.reverse()
print(nums)
# count the number of times 3 appears in the list
counter=nums.count(3)
print  ("The number 3 appears ",counter,"times")
```

```
[1, 2, 3, 3, 3, 3, 4, 5, 6, 9, 11]

[11, 9, 6, 5, 4, 3, 3, 3, 3, 2, 1]

The number 3 appears  4 times
```

Lists can be sorted in numerical order and alphabetical order.

```
meats = ['beef', 'chicken', 'lobster', 'lamb']

#sorted returns this list, but sorted alphabetically

for meat in sorted(meats):

   print(meat)
```

```
beef

chicken

lamb

lobster
```

5. Counting items in a list
   list.count(item) will count the number of times item appears in the list.

6. Determining the length of a list
   len(list) will count the number of items in a list. For example :

```
nums = [3, 5, 6, 3, 2, 1, 9, 11, 3, 4,3]

print( "There are ",len(nums), "items")
```

```
There are 11 items
```